Socratic AI Tutor

Dataset Preprocessing Documentation

SocraTeach Multi-turn + MathDial • Colab workflow • JSONL export

Project objective	Prepare tutoring dialogues for supervised fine-tuning (SFT) of Qwen2.5-1.5B-Instruct.
Sources	SocraTeach Multi-turn and MathDial chat-format dataset.
Processed records	35,151 SocraTeach dialogues + 2,262 MathDial conversations = 37,413 candidate conversations.
Output format	UTF-8 JSON Lines (JSONL), one conversation per line, represented as a list of chat messages.

Scope note: this document records the preprocessing steps and outputs observed in the Colab workflow. The merged file is a candidate training corpus, not a claim that the dataset is fully optimized, deduplicated, or ready for final model training.

1. Purpose and intended use

The project aims to fine-tune an instruction-tuned language model to behave more like a Socratic tutor: ask guiding questions, respond to learner attempts, provide hints, and support step-by-step reasoning rather than immediately revealing a complete answer. The two source datasets provide English math-tutoring dialogue examples. They are a useful starting point for learning tutoring interaction patterns, but they do not by themselves cover the project's intended college-level machine-learning topics such as linear regression, gradient descent, and model training.

2. Source datasets

2.1 SocraTeach Multi-turn

Hugging Face: ulises-c/SocraTeach_Multi [1]. The dataset card describes 10,273 math problems from GSM8K and MAWPS, with multiple Socratic dialogues and simulated student personas. Each source row contains fields such as id, question, analysis, answer, steps, and dialogues. A dialogue turn may include the tutor text in `system`, a student reply in `user`, and a `user_type` label.

2.2 MathDial chat-format dataset

Hugging Face: eth-nlped/mathdial-chat [2]. The loaded dataset exposes train and test splits, with 2,262 training examples and 599 test examples. The training rows contain question, ground_truth, conversation, student_incorrect_solution, and student_profile. The conversation field is already a list of role/content messages, including system, assistant, and user roles.

3. Processing environment and tools

- Environment: Google Colab notebook.
- Library: Hugging Face Datasets for loading the datasets.

- Serialization: Python json module; UTF-8 encoding.
- Output format: JSONL, one JSON object per line, with a messages array.
- Target model considered for later SFT: Qwen/Qwen2.5-1.5B-Instruct [3].

4. Preprocessing procedure

4.1 SocraTeach processing

- Loaded the train split containing 10,273 source examples.
- Iterated over each source example and expanded its dialogues list into individual dialogue records.
- Mapped each non-empty source system field to a Qwen-style assistant message because the field contains the tutor's utterance, not a global system instruction.
- Mapped each non-empty source user field to a user message.
- Preserved turn order, including the final tutor message when it appeared in a turn marked END=True.
- Kept dialogues that contained both user and assistant roles; no dialogues were skipped in the observed run.
- Serialized each dialogue to JSONL as {"messages": [...]} for the standalone processed file.

4.2 MathDial processing

- Loaded the train split only (2,262 examples); the 599-example test split was not processed into the training file.
- Read the existing conversation list and retained valid system, user, and assistant roles.
- Trimmed leading/trailing whitespace and removed empty or whitespace-only message contents.
- Kept all 2,262 conversations; four empty messages were removed and no conversations were skipped.
- Preserved the system message as a system message, since it provides scenario context and tutor instructions.
- Serialized each conversation as one JSONL line with a messages array.

4.3 Combining the datasets

The merge cell appends the processed SocraTeach dialogues and processed MathDial conversations. It adds a source metadata field with value `soctrateach` or `mathdial`, alongside the messages list, to support provenance tracking. The combined file name in the supplied code is `socratic_tutor_combined.jsonl`. The merge cell checks that records have assistant messages and writes one JSON object per line.

Training-format caution: the combined file contains a top-level source field as well as messages. Before passing it to an SFT trainer that expects only conversational messages, load each row and select the messages field, or write a training-only JSONL containing only {"messages": [...]} records.

5. Observed validation results

5.1 SocraTeach

Metric	Observed result
Original source rows	10,273
Extracted dialogues	35,151
Messages after conversion	336,051
Assistant messages	185,600
User messages	150,451
Missing tutor/system text occurrences	2
Skipped dialogues	0
Empty messages in converted records	0
Invalid records	0
Duplicate dialogue IDs	0
JSON serialization check	Passed

5.2 MathDial

Metric	Observed result
Training examples	2,262
Test examples held out	599
Processed conversations	2,262
Empty messages removed	4
Invalid-role messages removed	0
Conversations skipped	0
Exported JSONL lines	2,262
Validation	Passed

Expected combined count if the merge cell is run successfully: 35,151 + 2,262 = 37,413 conversations. This is an arithmetic expectation from the two processed lists; the user has not supplied a merge-cell output confirming the combined file was created or downloaded.

6. Data format examples

Standalone processed record (simplified)

```
{
  "messages": [
    {"role": "assistant", "content": "What do you think the first step is?"},
    {"role": "user", "content": "Maybe multiply the length and width?"},
    {"role": "assistant", "content": "Good start. What would that product represent?"}
  ]
}
```

Combined record structure

```
{
  "source": "mathdial",
  "messages": [
    {"role": "system", "content": "You are a friendly and supportive teacher."},
    {"role": "assistant", "content": "What do you think the first step is?"},
    {"role": "user", "content": "Maybe multiply the length and width?"}
  ]
}
```

7. Limitations and recommended next steps

- Domain coverage: both sources are math-focused. Add a curated, quality-reviewed dataset for college-level machine-learning topics before claiming the tutor can teach linear regression, gradient descent, or other MLT content.
- Conditioning information: the SocraTeach source includes question, steps, and user_type/persona information. The initial preprocessing described here drops those fields from the messages. The SocraTeach dataset card states that steps and user_type are important conditioning signals [1]. For a stronger training dataset, preserve them in an explicit system/developer-style context or redesign the examples so they are not lost.
- Conversation starts: some SocraTeach dialogues begin with a tutor question, without a preceding user message. This reflects the source dialogue structure. Consider adding a clear problem/context message before fine-tuning, or document this as a limitation.
- Supervised fine-tuning loss: ensure the training pipeline computes loss on assistant outputs rather than training the model to imitate student/user messages. Use the tokenizer's chat template and the trainer's assistant-only loss support where available.
- Evaluation: create a held-out evaluation set and measure tutoring quality (does it ask a useful next question, respond correctly to misconceptions, avoid giving away the answer too early, and remain mathematically correct?). Do not train on MathDial test examples.
- Deduplication: the current workflow checks duplicate dialogue IDs for SocraTeach, but does not perform semantic or cross-dataset duplicate detection.
- Licensing: check source licences and attribution requirements before redistributing records or derivatives. The SocraTeach card identifies CC BY-NC 4.0 [1]. MathDial's Hugging Face dataset variants and original repository should be checked directly because licence metadata can differ across distributions [2, 4]. This documentation is not legal advice.

8. Reproducibility and expected files

File	Purpose
soctrateach_processed.jsonl	Processed SocraTeach dialogue records; one messages array per line.
mathdial_processed.jsonl	Processed MathDial training conversations; system/user/assistant roles retained.
socratic_tutor_combined.jsonl	Combined candidate corpus with source metadata and messages, if merge cell is ru

The notebook cells shown in the project conversation are the operational record for this run. For reproducibility, commit the exact preprocessing notebook/script, library versions, validation output, and dataset source URLs alongside the documentation. Do not upload data files publicly until licence review is complete.

9. References

[1] Ulises Chavarria, “SocraTeach_Multi” dataset card, Hugging Face. https://huggingface.co/datasets/ulises-c/SocraTeach_Multi. Accessed 29 October 2026. Dataset card lists CC BY-NC 4.0 and requests citation of the SocraticLM paper.

[2] ETH Zurich, Image Language, Reasoning and Education Lab, “mathdial-chat” dataset, Hugging Face. https://huggingface.co/datasets/eth-nlped/mathdial-chat. Accessed 29 October 2026.

[3] Qwen Team, “Qwen2.5-1.5B-Instruct” model card, Hugging Face. https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct. Accessed 29 October 2026. Model card lists Apache-2.0.

[4] Macina, J., Daheim, N., Chowdhury, S., Sinha, T., Kapur, M., Gurevych, I., and Sachan, M. (2023). “MathDial: A Dialogue Tutoring Dataset with Rich Pedagogical Properties Grounded in Math Reasoning Problems.” Findings of EMNLP 2023, pp. 5602–5621. Project repository.

[5] Liu, J., Huang, Z., Xiao, T., Sha, J., Wu, J., Liu, Q., Wang, S., and Chen, E. (2024). “SocraticLM: Exploring Socratic Personalized Teaching with Large Language Models.” NeurIPS 2024. Paper PDF.

Suggested citation note for a project report: cite the original SocraTeach/SocraticLM paper and the MathDial paper, as well as the exact Hugging Face dataset cards used to download the processed distributions.